

JIMMY LIU

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PROFESSIONAL SUMMARY

Technical Program Manager with 7+ years running ML data operations: 5M+ labeled frames across 20+ production pipelines and 10 perception DNNs, coordinated across the US, China, India, and Vietnam. Deep familiarity with managing in-house, external vendor, and specialist-provider sourcing for data – evaluating across cost, speed, and data accuracy. Stands up the new quality-review and audit systems that keep vendor output honest. Built an ML feedback loop raising labeling quality from 30% to 95%+, and an evidence-first release workflow (300+ releases, zero rollbacks) that makes data defensible to model teams.

EXPERIENCE

NVIDIA | Santa Clara, CA

Technical Program Manager, ML Operations

January 2024 – Present

- Owned end-to-end ML data delivery across 20+ production pipelines and 10 perception DNNs, coordinating global vendor and operations teams across five time zones (US, China, India, Vietnam) to deliver 5M+ labeled frames.
- Built an end-to-end ML feedback loop for operations teams, raising labeling output quality from 30% to 95%+, with quality reporting, audit workflows, escalation paths, and a data flywheel for model improvement.
- Led active-safety vendor sourcing strategy for AEB scenarios: compared in-house, track-fastest vendor, and specialist-provider paths across cost, speed, execution complexity, and expected data accuracy; found the fastest path risked technically complete but unusable safety data. Recommended a capability-based model (specialist provider DRI + retained fallbacks) with an ML engineer onsite to catch invalid runs before ingestion; organization adopted the model.
- Authored a multi-region operating model with clear SLAs, escalation paths, formal Jira intake, and cross-functional reporting dashboards; increased labeler throughput by 40%.
- Built an evidence-first dataset-release workflow using source-aware SQL, deterministic placement rules, deduplication, changelogs, and human review gates; moved cadence from weekly to daily with 300+ releases and zero rollbacks.
- Led R18 Traffic Sign recovery: refined initial engineering task, preventing an oversized 300+ labeler staffing pull; built a two-week surge plan scaling to 230 labelers with ROI labeling, delivering approximately 60,000 frames.
- Owned the L3 data-loop delivery program with a bottoms-up demand-and-capacity model against a 1.1M-clip requirement; delivered approximately 500,000 unique clips and supported a lidar-perception MVP.
- Spearheaded AI enablement for a 9-person MLOps TPM team; built self-service analytics and evidence-to-Jira tooling reducing cycle times from days to minutes.

Technical Program Manager, AV Data Operations

May 2022 – January 2024

- Led end-to-end delivery of 13,000+ hours of driving data across 11 vehicles and six geographic regions with a cross-functional team of 22 operators.
- Drove roadmap and execution for Active Safety dataset delivery supporting AEB and LSS features, contributing to a 5-star NCAP rating for the 2026 Mercedes-Benz CLA; primary data-delivery contact across Mercedes and NVIDIA for 1,000+ engineers.
- Defined operational-readiness gates – vehicle stability, operator usability, downstream data consumability – converting a one-time checkpoint into an end-to-end contract spanning ingestion, curation, labeling, release, and model consumption.

- Led DataOps operational readiness for Hyperion 8.1 bring-up across 8 successive validation builds (Blue MVP through Green-3.2.3, non-production lidar/sensor sets): established three readiness gates – engineering stability, operational usability, downstream consumability – that redefined engineering's handoff criteria; challenged premature readiness by returning failing builds; delivered pilot data to downstream MLEs to validate sensor-data quality; ran daily stakeholder reporting to hold alignment.
- Defined P50/P90 uptime KPIs, improving data yield 10%+.

Magic Leap | Sunnyvale, CA

Senior Data Collection Engineer

August 2021 – May 2022

- Owned the Sunnyvale data collection process end-to-end as subject-matter expert for ground truth, motion capture, and lidar mapping systems, supporting four algorithm teams.
- Redesigned priority data-collection pipeline, improving efficiency 40%.
- Built Python automation for prototype-device configuration collection, reducing preparation time 10%.
- Developed parallel-processing pipeline converting raw binary data to customer-ready format with Python, Git, Docker, and Google Cloud Platform.

Amazon Lab126 | Sunnyvale, CA

Data Collection Specialist

June 2020 – August 2021

- Led multi-project data-collection programs from pilot through operational launch.
- Developed SQL automated-tagging workflows for Alexa Q&A, increasing delivery speed 30% and saving 8+ hours weekly.
- Built SQL and ETL pipelines for computer-vision data, improving quality through systematic tag corrections.

TECHNICAL SKILLS

Data Operations: Global vendor sourcing and selection, SOW and delivery tracking, quality assurance and audit systems, SLA/SLO management, labeler throughput and surge planning, multi-region operations (US/CN/IN/VN), escalation management

Eval & Release Quality: ML feedback loops, human-in-the-loop review gates, evidence-first dataset release, source provenance, deduplication, changelogs, review-gated and fail-closed workflows, quality dashboards

Technical Fluency: Python, SQL/Trino, ETL, Jira APIs, Redash, YAML, Git/Gerrit, Docker, Google Cloud Platform, perception systems (camera/radar/lidar), ADAS, ground-truth pipelines

EDUCATION & CERTIFICATIONS

University of California, Santa Barbara

Bachelor of Science in Biological Sciences | Graduated 2018